

Khatabook Growth Operations & Decision Playbook

Executive strategy briefing covering attribution boundaries, Simpson's paradox, and merchant loop dynamics across Tier 2/3 Indian retail digitization.

1. Commercial Diagnostics & Performance Metric Standards

1.1 Aggregation Integrity: Ratio-of-Sums vs. Row-Level Averages

Ad networks report performance metrics across fragmented ad sets with orders-of-magnitude volume disparities. In Campaign 1178, true Ratio-of-Sums CTR is **0.0176%** (36,068 clicks / 204,823,716 impressions), whereas unweighted row-averaging yields **0.0162%** (-7.9% understated). Across all 1,143 ads, unweighted row-averaging yields **0.0164%** vs true ratio-of-sums **0.0179%** (-8.2% understated). Averaging row-level rates assigns equal weight to micro-budget ad sets with small, noisy sample sizes.

The Growth Rule: All portfolio and segment-level conversion rates, CTRs, and cost metrics must be strictly aggregated as $SUM(Numerator) / SUM(Denominator)$. Row-averaged rates are prohibited in executive reporting.

2. Mathematical Variance Attribution (The 3-Factor Efficiency Bridge)

When comparing **Campaign 936** (\$2,893.37 across 464 ad sets) to **Campaign 1178** (\$55,662.15 across 625 ad sets, a 19.2x media budget expansion), Cost per Approved Conversion escalated from **\$15.81** to **\$63.83** (+303.7% cost inflation). Decomposed via sequential substitution attribution:

Lever	Observed Shift	Dollar Impact	Commercial Diagnosis
Lever 1: CPM Effect	\$0.356 -> \$0.272 (-23.6%)	-\$3.74	Auction efficiency improved at scale; cheaper inventory.
Lever 2: CTR Decay	0.0244% -> 0.0176% (-27.9%)	+\$4.66	Creative resonance decay; audience reach fatigue (+9.7%).
Lever 3: Conv Cliff	9.22% -> 2.42% (-73.8%)	+\$47.10	Post-click form collapse; accounted for 98.1% of cost surge.
Total Explained	Net Gap: +\$48.022	+\$48.02	Reconciliation Residual = \$0.00000000 (Exact Identity)

3. Confounding Control via Direct Standardization

Direct standardization re-weights both campaigns against an identical pooled benchmark distribution across all 4 age categories. **Campaign 936** adjusted unit cost shifts from \$15.81 to \$23.69, while **Campaign 1178** shifts from \$63.83 to \$78.16. The **3.3x efficiency gap remains completely intact** under constant demographic weighting, proving that audience mix shift is NOT the root cause.

4. Multi-Stage Funnel Transition Diagnostics

Funnel Transition Stage	Camp 936	Camp 1178	Drop-Off Variance	Diagnosis
Stage 1 -> 2: Impression -> Click	0.0244%	0.0176%	-27.9%	Creative resonance decay
Stage 2 -> 3: Click -> Enquiry	27.07%	7.40%	-72.7% (92.6% Bounce)	CRITICAL BOTTLENECK: Form friction
Stage 3 -> 4: Enquiry -> Approved	34.08%	32.67%	-4.1% (Near Parity)	Lead qualification parity maintained

5. Khatabook Merchant Unit Economics & Payback Crossover

Khatabook operates three interconnected growth loops: (1) Paid vernacular Meta/Google ads, (2) Organic WhatsApp Udhaar payment reminders (K-factor: 0.30–0.45), and (3) Fintech monetization (■125/mo Soundbox SaaS + UPI interchange + 2.5% loan origination spread). A **10% relative improvement in D1 merchant activation** reduces fully loaded merchant CAC by **~28%** without extra ad spend, driving payback breakeven in **4.2 to 5.8 months**.